

RESEARCH ARTICLE

Adaptive Throat-Sieving Gate in a Metal Azolate Framework Enabling Synergistic Equilibrium-Kinetic Separation of Propylene and Propane

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ABSTRACT

Developing porous adsorbents for propylene/propane (C₃H₆/C₃H₈) separation faces the challenge of integrating high adsorption capacity with fast adsorption kinetics. Herein, we address this challenge through precise pore control in a slightly flexible metal azolate framework NUM-27a, enabling synergistic equilibrium-kinetic separation of C₃H₆/C₃H₈. The periodically expanded throat gate enables effective impeding the diffusion of C₃H₈, while a large pocket-shaped cavity decorated with exposed oxide groups facilitates exceptional C₃H₆ capture at low pressures. Specifically, NUM-27a achieves exceptional low-pressure C₃H₆ capture (89.61 cm³ cm⁻³ at 0.1 bar) and a record C₃H₆ packing density (310.0 g L⁻¹ at 0.01 bar). Kinetic analysis further reveals effective diffusion coefficient for C₃H₆ is 1.57 × 10⁻⁴ s⁻¹ at 298 K. Gas-loaded single crystal X-ray diffraction analysis coupled with computational simulations elucidate that the intrinsic pore geometry underpins the unique adsorption and separation performance. Breakthrough experiments validate the outstanding separation performance of NUM-27a for C₃H₆/C₃H₈ mixtures. Moreover, the great stability, recyclability, and low-cost precursors of NUM-27a underline its potential as a reliable adsorbent for C₃H₆/C₃H₈ separation. The work unveils the adaptive throat-sieving gate strategy with optimal separation performance for challenging gas separations.

1 | Introduction

Propylene (C₃H₆) stands as the primary olefin feedstock in the petrochemical industry, with the global demand for C₃H₆ projected to reach 170 million tons in 2025 [1, 2]. The pivotal role of C₃H₆ in the petrochemical industry is underscored by its indispensable position in meeting the surging demand for key chemicals, including but not limited to polypropylene, acrylonitrile, and isopropanol [3, 4]. In current industrial practice, C₃H₆ is predominantly produced through naphtha steam cracking or gas oil catalytic cracking, with propane (C₃H₈) concurrently

generated as a major by-product [5, 6]. The C₃H₈ impurity can severely impact the polymerization of C₃H₆, making its separation essential for achieving polymer-grade purity (>99.5%) [7, 8]. The conventional separation of C₃H₆/C₃H₈ mixture relies on a cyclic distillation-compression process conducted in massive splitter towers, which can reach heights of up to 300 feet and house more than 200 trays—under operating conditions of approximately 243 K and 0.3 MPa [5]. Evidently, such a separation process remains one of the most capital-intensive and energy-demanding operations in the industry [9, 10]. Therefore, the development of energy-efficient separation

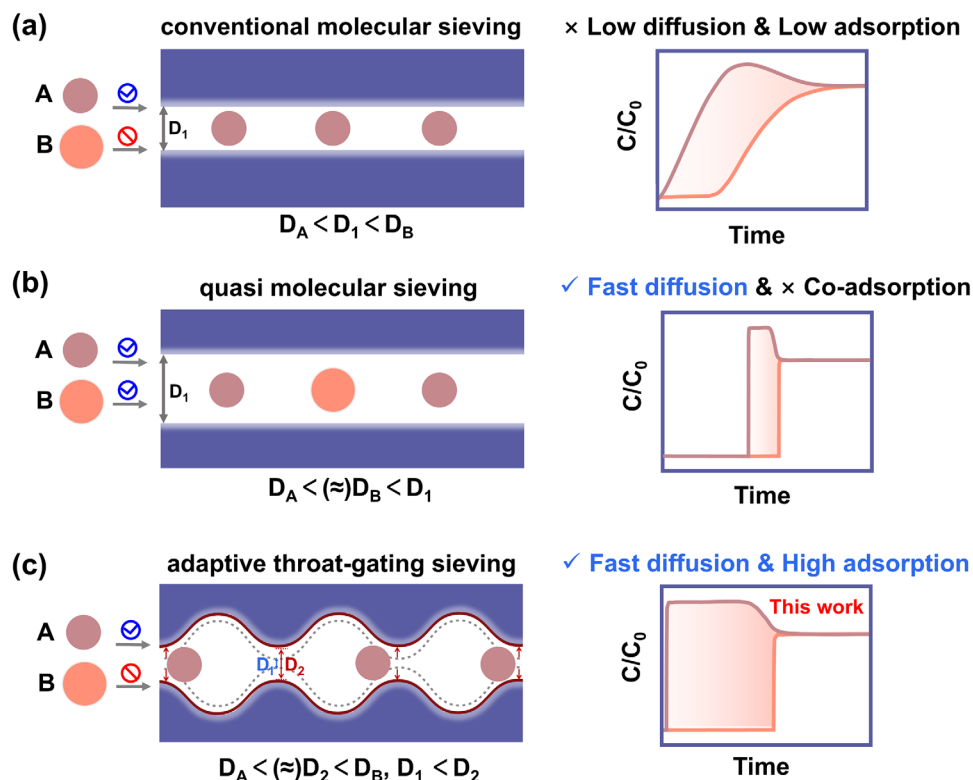


FIGURE 1 | Schematic illustration of sieving behaviors in different pore systems. (a) Conventional molecular sieving occurs when the target molecule A diffuses in a single narrow channel, and the impurity molecule B is completely blocked. (b) Quasi-molecular sieving occurs when target molecule A and impurity molecule B exhibit discrepant diffusion rates while permeating through the narrow channels. (c) Adaptive throat-gate sieving, where the expanded throat-gate acts as a sieve, blocking the entry of impurity molecule B. The adjacent pocket-shaped space provides an ample “adsorption domain” to accommodate target molecule A. Rose pink ball: target molecule A, the diameter is D_A ; Coral orange ball: impurity molecule B, the diameter is D_B .

technologies, such as adsorptive separation using porous materials, is crucial for potentially reducing the cost and the energy consumption [11].

Developing specialized adsorbents is the key to the success of adsorptive technologies [12, 13]. Among the emerging adsorbent materials, metal-organic frameworks (MOFs) have been extensively explored as promising candidates for challenging gas separations, owing to the high tunability and versatile functionality at the molecular level [14–30]. Particularly, molecular sieving with infinite selectivity is recognized as the most ideal separation approach [4, 31–39]. Several sieving MOFs with tailored pore size and shape to selectively exclude the larger C_3H_8 molecules from the binary C_3H_6/C_3H_8 mixtures have been reported [3, 5, 40–45]. The conventional molecular sieving approach mostly relies on single narrow channels that allow the complete separation of C_3H_6 from C_3H_8 based on molecular size or shape cut-off, avoiding the co-adsorption of C_3H_8 (Figure 1a). However, since the kinetic diameter difference of C_3H_6 and C_3H_8 is less than 0.4 Å, precise regulation of the pore size/shape matching is very challenging [46]. In addition, this approach typically restricts the diffusion rate and diminishes the host-guest interaction between the framework and C_3H_6 due to limited pore volume [22, 47, 48]. On the other hand, quasi-molecular sieving has been proven beneficial by accelerating C_3H_6 diffusion within the channel, while providing an extended adsorption domain to accommodate high C_3H_6 loading (Figure 1b) [46, 49, 50]. Yet it suffers from the

drawback of co-adsorption of C_3H_8 , which further reduces the separation selectivity and the actual work production capacity. Thus, it is particularly challenging to optimize the adsorption kinetics and host-guest interaction for sieving materials because of the difficulty of precise pore configuration control at the sub-angstrom scale.

In light of this, a highly promising strategy for the challenging C_3H_6/C_3H_8 separation involves developing a pore-engineered framework that integrates both an adaptive throat sieving gate and large cavities. In this rational design, the throat sieving gate exhibits conformational adaptation to C_3H_6 , with its structural flexibility enabling subtle adjustments in aperture size and geometry upon interaction with C_3H_6 molecules. These adjustments optimally match the dynamic diameter of C_3H_6 while remaining a steric barrier to the larger C_3H_8 (Figure S1). The adjacent large cavities provide an ample “adsorption domain”, where internal volume not only accommodates a high loading of C_3H_6 molecules but also minimizes diffusion resistance to allow the adsorbed C_3H_6 to migrate freely (Figure 1c). This integrated pore architecture endows distinct thermodynamic and kinetic behaviors: the smaller C_3H_6 effortlessly permeates the gate for rapid diffusion, while engaging in strong host-guest interactions with the exposed functional groups within the aperture. In contrast, C_3H_8 is sterically blocked by the throat-sieving gate (kinetic exclusion) and thus excluded from the framework. This synergistic interplay between equilibrium and kinetics can

overcome the long-standing trade-off between adsorption capacity and diffusion dynamics in sieving separation.

With this in mind, we present a new and micro-flexible MOF, $\text{Zn}_2(\text{ATZ})_2(\text{SO}_4)$ (ATZ = 3-amino-1,2,4-triazolate), termed NUM-27a, which features a periodically porous configuration that aligns with the “adaptive throat-sieving gate” model. As hypothesized, the narrow throat gate effectively impedes the diffusion of C_3H_8 and the pocket-shaped space, which provides sufficient oxygen affinity sites for C_3H_6 . NUM-27a shows an ultra-high ideal adsorbed solution theory (IAST) selectivity for equimolar $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixtures (exceeding 3×10^4). Additionally, kinetic analysis using an effective diffusion coefficient (D') reveals fast adsorption kinetics for C_3H_6 . Breakthrough experiments confirm the successful separation of a binary $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixture as well as excellent cycling stability over multiple cycles of tests. Gas-loaded single crystal X-ray diffraction (SCXRD) analysis and modeling simulations further provide valuable insights into the separation mechanism at the molecular level. These studies provide in-depth insights into the unique interactions and diffusion dynamics within the framework, furnishing a comprehensive understanding of its remarkable sieving separation behavior.

2 | Result and Discussion

2.1 | Synthesis and Characterization

A new sulfate-pillared metal azolate framework, termed NUM-27 [$\text{Zn}_2(\text{ATZ})_2\text{SO}_4 \cdot 4\text{H}_2\text{O}$], was synthesized for the first time by the hydrothermal reaction of zinc sulfate heptahydrate and ATZ in a mixed solution of water and trifluoroacetic acid at 453 K for 48 h. The roles of solvent and reaction temperature are crucial, as using more commonly used solvent systems or lower reaction temperatures would form different products, such as HAF-1 [6] and $\text{Zn}_2(\text{aTz})_2\text{SO}_4$ [51] (Figures S2, S3, and Table S1). Single crystal X-ray diffraction analysis demonstrates that NUM-27 crystallizes in the *Pc* space group (Table S2). Specifically, its simplest asymmetric unit consists of two Zn^{2+} , two ATZ ligands, and one SO_4^{2-} . Each Zn^{2+} center is coordinated with three nitrogen atoms from three ATZ ligands and one oxygen atom from the SO_4^{2-} , adopting a four-coordinate geometry (Figure S4). Two Zn^{2+} centers are bridged by two ATZ ligands to form a [$\text{Zn}_2(\text{ATZ})_2$] secondary building unit (SBU). The adjacent [$\text{Zn}_2(\text{ATZ})_2$] SBU undergoes a rotation of approximately 90° , positioning itself in two mutually perpendicular planes. This rotational effect causes the out-of-plane distortion of ATZ rings, which in turn drives the coordinated deformation of Zn–N coordination bonds, constructing a wavy 2D [$\text{Zn}_2(\text{ATZ})_2$] $_n$ layer. When these layers stack closely together, SO_4^{2-} serves as a bridge, facilitating the formation of a distinct 3D framework (Figure S5). Notably, immersing NUM-27 in MeOH for several minutes will trigger a distinct single-crystal-to-single-crystal transformation. The new phase, termed NUM-27-MeOH [$\text{Zn}_2(\text{ATZ})_2\text{SO}_4(\text{MeOH}) \cdot \text{MeOH}$], crystallizes in the monoclinic *P2*₁ space group. Structural comparison reveals that Zn^{2+} centers exhibit two distinct coordination geometries: half of the Zn^{2+} centers coordinate with MeOH molecules, forming a distorted tetrahedral geometry, while the other half adopt a distorted trigonal bipyramidal geometry in NUM-27-MeOH (Figures S6–S8). However, NUM-27-MeOH is not a stable final phase. The unbalanced coordination environment

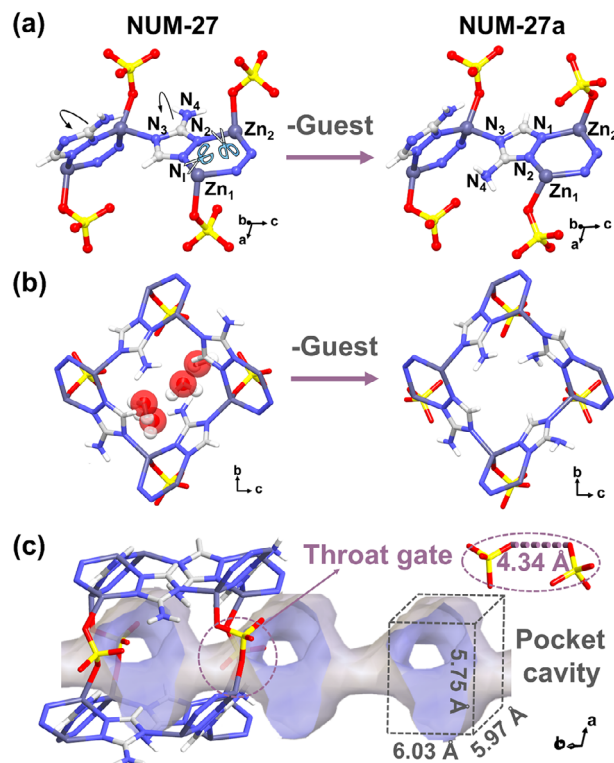


FIGURE 2 | Illustration of the framework structure and pore geometry of NUM-27 and NUM-27a. (a) The single-crystal-to-single-crystal transformations. (b) Framework of NUM-27 and NUM-27a. (c) Pore channels of NUM-27a with a large pocket cavity and a narrow throat gate viewed along a representative direction. Color code: C, gray; N, light blue; H, white; O, red; S, yellow; Zn, blue-gray.

formed by the involvement of MeOH molecules within the framework serves as a thermodynamic driving force, further triggering structural rearrangement. Upon guest removal, the cleavage of Zn–N coordination bonds occurs, accompanied by the conformational rotation of ATZ rings driven by thermodynamics. Ultimately, the coordination mode between Zn^{2+} centers and ATZ rings is completely reconfigured, and MeOH molecules dissociate from the coordination environment, leading to the formation of the activated phase NUM-27a [$\text{Zn}_2(\text{ATZ})_2\text{SO}_4$] (Figure 2a,b, and Figure S9). The skeleton of NUM-27a incorporates a regular pocket-shaped space interconnected by a narrow throat-sieving gate (Figure 2c). Notably, the throat gate aperture is constructed by four Zn^{2+} centers, two ATZ rings, and two SO_4^{2-} , with the shortest distance between adjacent SO_4^{2-} being 4.34 Å. The pocket-like cavity ($5.97 \times 6.03 \times 5.75 \text{ Å}^3$) possesses adequate volume to accommodate both C_3H_6 and C_3H_8 guest molecules, whereas the throat gate size (4.34 Å) is slightly narrower than C_3H_8 . Moreover, the degrees of freedom provided by the tridentate ATZ rings and SO_4^{2-} groups facilitate the deformation of the wavy 2D [$\text{Zn}_2(\text{ATZ})_2$] $_n$ layer when adapting to external changes, giving the structure a certain extent of flexibility to allow target molecules to pass through.

From the optical microscope image (Figure S10), it is distinctly seen that NUM-27 exhibits a colorless block. Powder X-ray diffraction (PXRD) analysis verified the phase purity of the NUM-27 and NUM-27a (Figures S11 and S12). However, certain diffraction

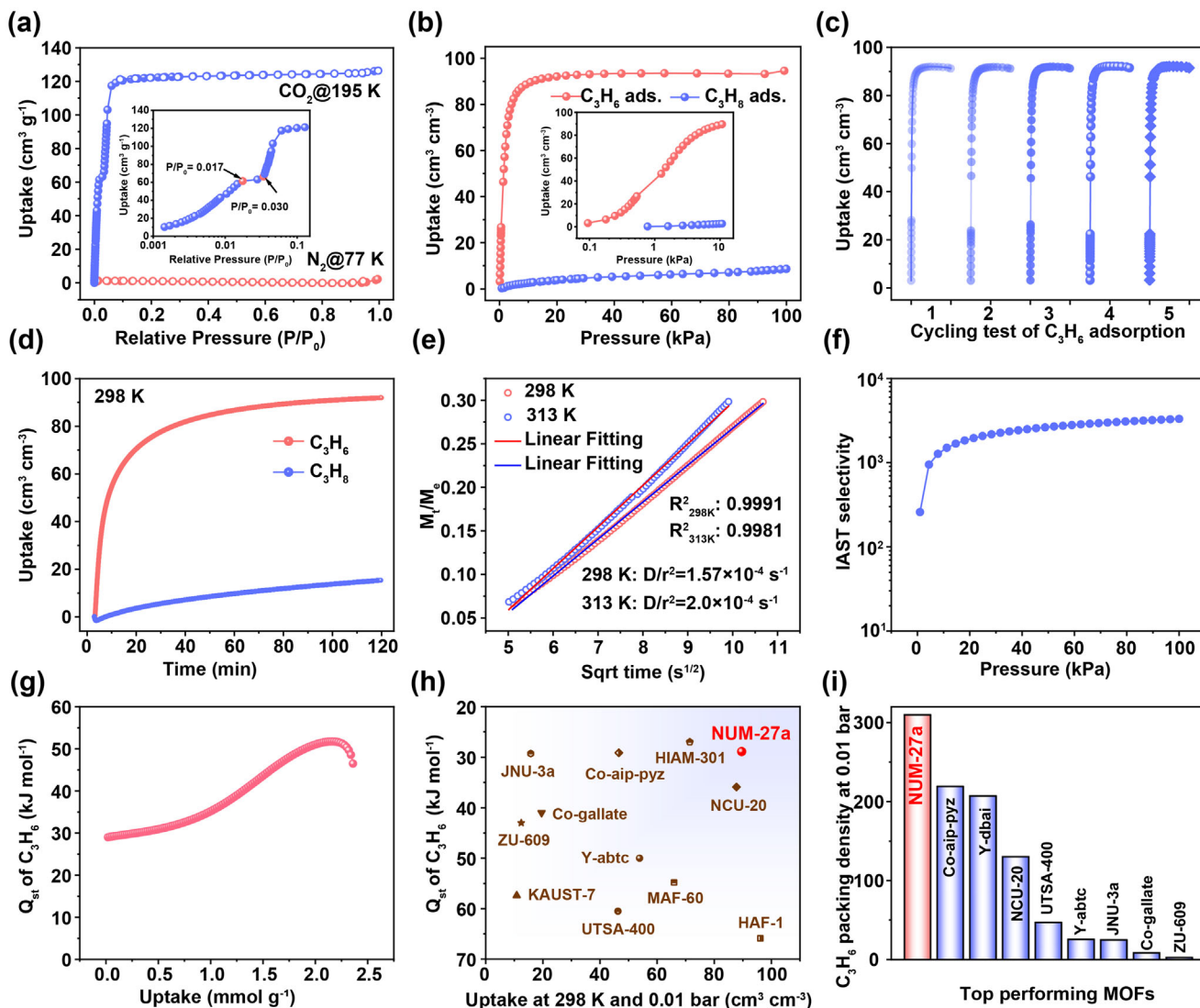


FIGURE 3 | Gas sorption properties of NUM-27a. (a) N_2 adsorption–desorption isotherm at 77 K and CO_2 adsorption–desorption isotherm at 195 K for NUM-27a. (b) The adsorption isotherms of C_3H_6 and C_3H_8 on NUM-27a at 298 K (Inset: the isotherm is presented in semilogarithmic scale to point out details in the low-pressure region). (c) Cycling test of C_3H_6 adsorption measurements on NUM-27a at 298 K from 0 to 1.0 bar. (d) The adsorption kinetic profiles of C_3H_6 and C_3H_8 on NUM-27a at 298 K. (e) The fitting of the kinetic curve for C_3H_6 at 298 and 313 K. (f) IAST selectivity for equimolar $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ mixture at 298 K. (g) Isothermic heat of adsorption of C_3H_6 for NUM-27a. (h) Comparison of the zero-coverage heat of adsorption of C_3H_6 and uptake at 298 K and 0.01 bar with those of typical MOF materials for $\text{C}_3\text{H}_6/\text{C}_3\text{H}_8$ sieving separation. (i) Comparison of C_3H_6 packing density at 298 K and 0.01 bar with those of top-performing MOF materials.

peaks (such as 100 and 011 planes) of NUM-27a exhibit a shift toward higher angles relative to those of NUM-27 (Figure 5a), indicating structural changes. The simulated PXRD patterns of HAF-1, $\text{Zn}_2(\text{aTz})_2\text{SO}_4$, and NUM-27a further corroborate the structural differences among these three frameworks (Figure S13). Thermogravimetric analysis (TGA) reveals an initial weight loss of 12.6% observed from 25°C to 171°C, which is attributed to the removal of four water molecules in NUM-27. This value is in agreement with the 15.5% weight loss calculated from the SCXRD data. Subsequently, a sharp and steep weight drop occurs upon further heating up to 400°C, indicating the onset of framework collapse. Notably, NUM-27a exhibits remarkable thermal stability, maintaining its structural integrity under argon atmosphere at temperatures up to 400°C (Figure S14). The permanent porosity of the NUM-27a was investigated using both N_2 and CO_2 as probe

gases (Figure 3a). Due to its narrowed channel, a significant diffusion barrier for N_2 at 77 K was observed in NUM-27a. As a result, CO_2 was chosen as the preferred probe. Notably, a subtle stepwise CO_2 adsorption behavior emerged in NUM-27a: at low relative pressures (ranging from near zero up to $P/P_0 = 0.017$), the CO_2 uptake increases gradually from approximately 10 to 60 $\text{cm}^3 \text{g}^{-1}$. Subsequently, a distinct plateau region emerges between $P/P_0 = 0.017$ and 0.03. Beyond this threshold ($P/P_0 > 0.03$), the uptake curve undergoes a steep, exponential ascent, reaching 120 $\text{cm}^3 \text{g}^{-1}$ at $P/P_0 = 0.1$. The Brunauer–Emmett–Teller (BET) surface area and pore volume are 502.88 $\text{m}^2 \text{g}^{-1}$ and 0.16 $\text{cm}^3 \text{g}^{-1}$, respectively, aligning closely with the theoretical value (Figures S15 and S16). And the pore size was estimated to be 4.58–6.28 Å (Figure S17). This sequential behavior reflects responsive structural dynamics of the framework, where initial low-pressure

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CARBON CAPTURE

A scalable metal-organic framework as a durable physisorbent for carbon dioxide capture

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Metal-organic frameworks (MOFs) as solid sorbents for carbon dioxide (CO₂) capture face the challenge of merging efficient capture with economical regeneration in a durable, scalable material. Zinc-based Calgary Framework 20 (CALF-20) physisorbs CO₂ with high capacity but is also selective over water. Competitive separations on structured CALF-20 show not just preferential CO₂ physisorption below 40% relative humidity but also suppression of water sorption by CO₂, which was corroborated by computational modeling. CALF-20 has a low enthalpic regeneration penalty and shows durability to steam (>450,000 cycles) and wet acid gases. It can be prepared in one step, formed as composite materials, and its synthesis can be scaled to multikilogram batches.

Capture of CO₂ after fossil fuel combustion requires CO₂ removal from a localized emission source but also regeneration and recycling of the capture system. Major challenges for the capture stage span materials design and development through to process engineering (1, 2). Flue gas has a low concentration of CO₂ diluted in mostly N₂ along with water and acid gases (3). Amine and solvent systems (4, 5) rely on contacting flue gas with a liquid that absorbs the CO₂ through a combination of chemical and physical absorption. Although CO₂ removal is effective, regeneration is energy intensive and can lead to chemical decomposition (6).

Solid sorbents represent a step-change technology for carbon capture (7–10) and have been demonstrated at smaller scales (11). Solids can bind CO₂ through either chemical or physical sorption (3, 7–10). In most cases, chemisorptive materials have higher capacity and selectivity for CO₂ (12). However, factors that enhance CO₂ binding often proportionally increase the energy needed to regenerate the sorbent and can enhance binding of competing gases. For the absolute CO₂ uptake, the relevant parameter

is working capacity under the operational cycling conditions to regenerate the solid sorbent (3). Selectivity over N₂ is typically reported, but sorption of CO₂ in the presence of water vapor is much less reported, especially for physisorptive capture systems (12–14). A physisorptive CO₂ capture solid would offer much lower regeneration costs, but it must have sufficient working capacity and selectivity in an actual flue stream in which gases are present with stronger intermolecular attractive forces than those of CO₂. Moreover, to translate to process productivity, the kinetics of sorption and release are as important as capacity.

Nearly all classes of porous solids have potential as solid sorbents for CO₂ capture (1–3, 7–10), including metal-organic frameworks (MOFs) (2, 3, 9, 12–14), in which chemical building blocks, pore sizes and shapes, surface functionalities, and even degrees of order can be varied to optimize CO₂ capture ability. More robust MOFs (15, 16), including ones that are stable in the presence of water (17–19) and steam (20), have been reported, although stability to wet acid gases is less common (21–23). For sorbent powder to be a usable material, it must be capable of formation in macroscopic shape for rapid mass transfer and thermal management, be durable in that form, and be available at scale (hundreds of thousands of tonnes) and reasonable cost (24, 25).

Solid sorbents optimized in an adsorption process have the potential to substantially decrease the CO₂ capture cost compared with traditional amine absorption processes because of lower regeneration energy, less chemical decomposition versus the solvent capture system, extensive use of stainless-steel owing to the corrosivity of amine solvents, and large plant footprint (4–6). Optimization of the solid sorbent

process must include high volumetric productivity in the presence of water (present in the flue gas) and the lowest regeneration energy. For regeneration, several processes are under evaluation, including vacuum swing, pressure swing, and temperature swing (26). Although cycling performance per sorbent volume or productivity is one of the main drivers of final CO₂ capture cost, there are several other parameters that affect operating and capital expenses of CO₂ capture. For solid sorbents, unlike solvent-based absorption, it is not feasible to continuously replace deactivated sorbents with fresh ones.

Here we present Calgary Framework 20 (CALF-20), a MOF with high capacity and selectivity for CO₂ despite a physisorptive mechanism and modest heat of adsorption. Its selectivity extends beyond N₂ to capture CO₂ in a wet gas. CALF-20 is exceptionally robust and stable to steam, wet acid gases, and even prolonged exposure to direct flue gas from natural gas combustion. Its single-step synthesis from commercially available components is highly scalable. The origin of the CO₂ philicity, despite CALF-20 being highly water resistant, was studied by simulation. Structuring of CALF-20 was performed, as well as competitive breakthrough experiments in wet gas streams that aligned with pure-component isotherms, heats of adsorption, and molecular modeling. In particular, not only can CALF-20 physisorb CO₂ up to and beyond 40% RH, but the presence of CO₂ actually suppresses water sorption. Finally, we present durability and CO₂ capture data on the MOF that are based on industrial testing.

Synthesis, structure, and gas sorption

CALF-20, [Zn₂(1,2,4-triazolate)₂(oxalate)], was initially prepared solvothermally and single crystals obtained through the in situ degradation of a dihydroxybenzoquinone derivative (see supplementary materials). CALF-20 is composed of layers of 1,2,4-triazolate-bridged zinc(II) ions pillared by oxalate ions to form a three-dimensional (3D) lattice and 3D pore structure (Fig. 1, A to C). Channels of 2.73 Å by 2.91 Å, 1.94 Å by 3.11 Å, and 2.74 Å by 3.04 Å along [100], [011], and [0 $\bar{1}$ 1], respectively (factoring van der Waals radii), that permeate the MOF result in a ~38% void volume. The one crystallographically unique Zn center is five-coordinate with a distorted trigonal bipyramidal geometry [Zn–O = 2.022(2), 2.189(3) Å; Zn–N = 2.007(2), 2.016(3), 2.091(3) Å]. The N atoms in the 1,2 positions of the triazolate bridge Zn dimers are linked to the next dimer by the N atom in the 4-position. The Zn coordination is completed by two oxygen atoms of a chelating oxalate group, and there are no open coordination sites. The bulk powder shows the same phase (Fig. 1D). Detailed structural analyses on pillared zinc triazolates have shown that layers can exist

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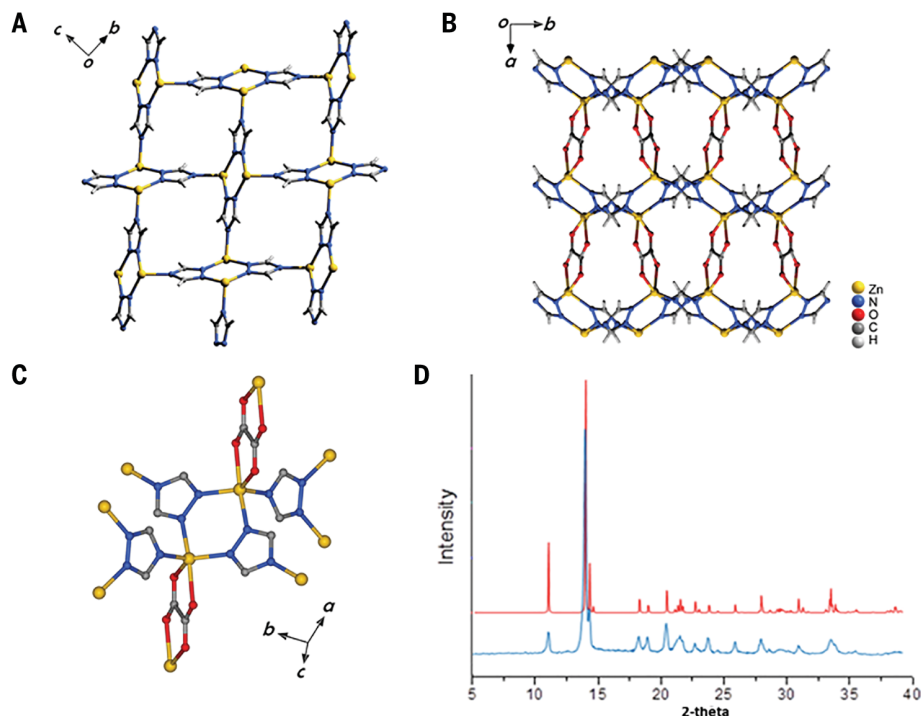


Fig. 1. Single-crystal structure of CALF-20. (A) View of the two-dimensional zinc triazolate grid. (B) View orthogonal to (A) showing the pillaring of the zinc triazolate layers by oxalate anions. (C) View of the zinc coordination sphere (H atoms removed). (D) Powder x-ray pattern simulated from the single-crystal structure (top) and obtained experimentally.

in different manifestations with varying degrees of buckling (27, 28). Indeed, since a provisional patent application was filed in 2014, a hydrated form of $[\text{Zn}_2(1,2,4\text{-triazolate})_2(\text{oxalate})]$ has been reported (29). This structure has the same connectivity but slightly different unit cell and pore dimensions. The specific pore structure affects sorption properties, and modeling was carried out with our obtained crystal data.

Gas adsorption experiments were performed for CO_2 and N_2 (Fig. 2A). The Langmuir surface area calculated from the N_2 isotherm at 77 K was $528 \text{ m}^2 \text{ g}^{-1}$, and the uptake for CO_2 was 4.07 mmol g^{-1} at 1.2 bar and 293 K. The zero-loading heat of adsorption for CO_2 was -39 kJ mol^{-1} (fig. S6), and the calculated selectivity for CO_2/N_2 by ideal adsorbed solution theory was 230 for a 10:90 CO_2/N_2 mixture. CALF-20 structured readily as a 20% polysulfone composite and retained the expected porosity (Fig. 2, B and C, and fig. S5). For CO_2 capacity and selectivity over N_2 as metrics, there are numerous other materials with noteworthy performance (30–36). The water sorption profile of CALF-20 was unusual in that, for a solid with good physisorptive capacity for CO_2 , it exhibited poor water uptake at low partial pressures (Fig. 2, D and E). Comparisons to zeolite 13X (37), as well as two other water-resistant MOFs, CAU-10 (38) and Al fumarate (39), are included in Fig. 2 and fig. S16. Moreover, higher-temperature

water isotherms showed that water uptake decreased more readily at higher temperatures than did the corresponding CO_2 isotherms.

Binding-site modeling

To gain insights into the nature of CO_2 binding in CALF-20 and its unusual water sorption behavior, we performed atomistic grand canonical Monte Carlo (GCMC) simulations (see supplementary materials). The experimental and simulated CO_2 and N_2 isotherms were in excellent agreement (see supplementary materials). Probability distributions of the guest molecules within the MOF allowed us to identify binding sites. The most probable CO_2 binding, which lies in the middle of the CALF-20 pore (Fig. 3A), had a binding energy of $-34.5 \text{ kJ mol}^{-1}$ based on the GCMC force field; the density functional theory (DFT) value with dispersion corrections was $-36.5 \text{ kJ mol}^{-1}$. The interatomic distances displayed were consistent with physisorption; the shortest distance was $3.03 \text{ \AA}$ between the CO_2 oxygen and a hydrogen of the triazole (fig. S8). Analysis of the binding energy revealed that the CO_2 -MOF interaction was dominated by attractive dispersion interactions (85%), with electrostatics contributing the balance.

Water adsorption isotherms are more challenging to simulate given the polar nature of water, which enables potentially strong interactions with the framework and with itself. The experimental water isotherm had a general

S-shape, where the water uptake was initially low until $\sim 10\%$ relative humidity (RH), at which point there was a steep rise until $\sim 30\%$ RH (Fig. 3B). These features indicated that water condensed in the pores, and they were reproduced in the simulated isotherm. After the initial steep rise in water uptake beyond 30% RH, the experimental isotherm showed a more gradual increase in adsorption until reaching a saturation limit at $\sim 11 \text{ mmol g}^{-1}$. However, for the simulated isotherm, the steep rise continued until full saturation at 40% RH and then flattened. The general S-shape and the saturation capacity of $\sim 11 \text{ mmol g}^{-1}$ were reproduced by the simulation.

A snapshot from the pure water simulation at 20% RH, where the water uptake was roughly half the saturation limit (Fig. 3C), revealed that the pores were either full of water molecules, forming a hydrogen-bonded network, or completely empty. In comparison, at 60% RH, where uptake had fully saturated, all the pores were full of hydrogen-bonded water molecules (fig. S9). The equilibrium distribution at 20% RH, where partially filled pores were not observed, suggested rapid condensation or evaporation of water. We extracted the water binding sites at 20% RH with the highest probability from the GCMC simulations, and the top three binding sites, in order, are labeled i, ii, and iii in Fig. 3D. The binding energies with the framework of the sites, -17.5 , -8.9 , and $-29.1 \text{ kJ mol}^{-1}$, respectively, were calculated by placing a single water molecule in the site with no other guest molecule present. The two most probable binding sites had a relatively low binding energy and were oriented away from the framework such that there were no hydrogen-bonding interactions with the oxalate linkers. Water molecules in these sites were poised to form hydrogen-bonding interactions with other water molecules, which suggested that the main driver for the initial water uptake was the interaction with other water molecules. This result was consistent with the experimentally observed water-uptake properties of CALF-20 at low RH.

Breakthrough studies

The intriguing CO_2 and water isotherms prompted a series of dynamic breakthrough studies (Fig. 4) on the CALF-20-polysulfone composite (see supplementary materials). Competitive CO_2/N_2 studies, with CO_2/N_2 mixtures of 5/95, 15/85, and 30/70, respectively (Fig. 4, A and B), confirmed the selectivity suggested by the pure-component isotherms. In the N_2 profiles, a sharp front, indicating complete breakthrough of N_2 , was observed at dimensionless time (ratio of experimental time to the time taken by a nonadsorbing tracer to travel through the column) $\bar{t} \sim 4$ in all three cases. The “roll-up” effect of N_2 , whereby the outlet composition of N_2 was higher than its inlet value, until CO_2 broke through is clearly

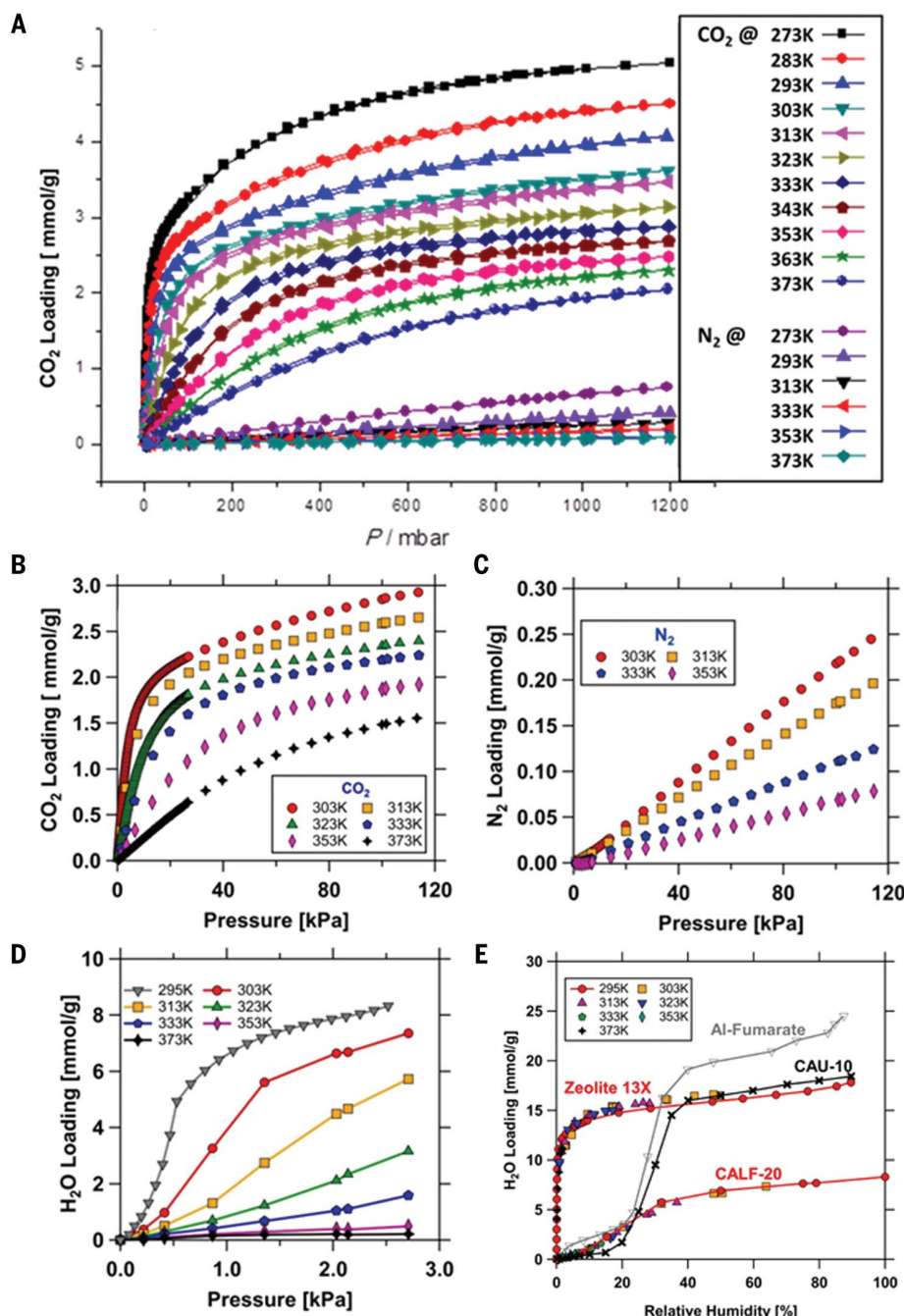


Fig. 2. Equilibrium gas uptake data on pure CALF-20. (A) CO₂ and N₂ isotherms from 273 to 373 K on pure CALF-20. (B to D) Structured CALF-20 (80% MOF:20% polysulfone). (B) CO₂ isotherms from 303 to 373 K. (C) N₂ isotherms from 303 to 353 K. (D) H₂O isotherms from 295 K to 373 K. (E) A comparison of H₂O isotherm on zeolite 13X (37), CAU-10 (38), Al fumarate (39) and structured CALF-20 at 295 K. The isotherms of CO₂ and N₂ were measured by volumetry, and that of H₂O was measured by gravimetry.

visible. The CO₂ concentration profiles showed different breakthrough times for various CO₂/N₂ compositions (Fig. 4A). Higher CO₂ composition in the feed led to shorter breakthrough times

We measured the competitive adsorption of CO₂ and H₂O using a combination of gravimetry, a study that measured the loading of CO₂ + H₂O by subjecting the sample to a moist

stream of CO₂ whose RH was controlled, and a breakthrough experiment that provided the competitive loading of H₂O in the presence of CO₂ (Fig. 4, B and C). The difference between the total loading from the gravimetry and the H₂O loading from the breakthrough provided the competitive CO₂ loading. Up to a value of 30% RH, the CO₂ loading was nearly unaffected (Fig. 4F), which was unexpected for a physisorptive

material but corroborated by the atomistic simulations. The CO₂ loading gradually decreased until it became negligible at RH > 80%. Additionally, the distinct shift in the H₂O isotherm in the presence of CO₂, compared to its pure-component isotherm, also confirmed the suppression of water sorption by CO₂.

To further demonstrate the physisorption of CO₂ by CALF-20 in wet environments, we measured the water breakthrough curves in air or CO₂ at two different RHs (Fig. 4, D and E). With the air experiment as a background, the water breakthrough was actually accelerated in CO₂, providing definitive support for the physisorptive preference of CALF-20 for CO₂ over water below 40% RH. The difference in water loading, exemplified by the area behind the breakthrough curve, between the two curves was pronounced. A comparison of both CO₂ and H₂O loading in competitive experiments (Fig. 4F and figs. S15 and S16) corroborated not only the sustained CO₂ capacity in wet gas but also the ability to suppress water sorption.

The nature of the water and CO₂ binding from the single-component water simulations and dry CO₂/N₂ simulations presented in Fig. 3 was consistent with the preferential binding of CO₂ over water observed at low RHs. Namely, CO₂ has strong binding sites in the center of the CALF-20 pores that precluded the formation of a hydrogen-bonded network that was responsible for the large uptake of water at high RHs. To corroborate this model, we performed multi-component simulations of CO₂, N₂, and water at varying RH. Figure S10 shows the comparison of simulated water uptake at various RHs from a single-component water simulation to that of multi-component simulations with 0.20 bar of CO₂ and 0.80 bar of N₂. The results were in good agreement with the experimental competitive isotherms shown in Fig. 4F. The simulations showed that without CO₂, water uptake at 20% RH is 6 mmol/g, whereas in the presence of CO₂, it was negligible. Only at 40% RH did water uptake reach 6 mmol/g when CO₂ was present. Calculated binding energies of most probable CO₂ and H₂O binding sites taken from the multi-component CO₂/N₂/H₂O simulations give $-17.5 \text{ kJ mol}^{-1}$ for H₂O and $-33.5 \text{ kJ mol}^{-1}$ for CO₂ (table S6). Calculated heats of adsorption, at zero loading and high loading (table S7), suggest that partially water-loaded pores were more attractive for subsequent water sorption than empty pores, and CO₂ had a stronger zero-loading heat of adsorption than water. A binding site analysis of the mixed CO₂/N₂/H₂O simulations is presented in the supplementary materials.

The low water-affinity yet CO₂-philic behavior of CALF-20 was enabled by its pore structure. Although a pore that is ideal for CO₂ is, of course, targeted in carbon capture, it is much less a focus that a pore be nonideal for water. Notably, a key feature of CALF-20 was the absence of any

strongly interacting functionality with CO₂. Although this property would be expected to moderate the affinity of the MOF for CO₂, the less specific dispersion interactions cumulatively compensate. As previously mentioned, dispersion interactions account for >85% of the binding energy in the most favorable CO₂ binding site. The boiling point of H₂O is 157°C higher than that of CO₂, so it is not expected that CO₂ would preferentially physisorb, but we can connect the competitive sorption in Fig. 4F with modeling in fig. S10. Interactions between guest molecules (40, 41), or in this case, the lack thereof as CO₂ blocked cooperative H₂O binding, could tip subtle balances in binding enthalpies. The pore itself is the critical element in performing a sorptive function (42). Other MOFs with low water affinity such as CAU-10 (38) and Al fumarate (39) have been reported and studied for CO₂ capture from wet gas. These MOFs have good CO₂/N₂ selectivity and reasonably low water affinity, as indicated by stepped water isotherms. However, CAU-10 loses CO₂ capacity above a RH value of 20%, and aluminum fumarate loses 17% CO₂ capacity at 14% RH (fig. S16). However, CALF-20 has a higher CO₂ capacity and retains it up to and beyond 40% RH. Also, neither shows the suppression of water sorption by CO₂ that is observed with CALF-20.

Flue gas sorption and scaling

Industrially, materials must absorb CO₂ from postcombustion flue gases at 100°C containing water vapor and acid gases, and endure stresses during regeneration as the sorbent goes through a temperature swing, pressure swing, or vacuum swing process. CALF-20 has been run through stability assessments from multiple academic, government, and industry partners and shows robust performance, as confirmed by retention of structure and gas adsorption properties. The retention of CO₂ capacity after being repeatedly heated to dry air at 150°C in the thermal gravimetric analysis (Fig. 5A) showed excellent stability (6, 43). This feature is key to high sorbent lifetime, as there is residual O₂ in the flue gas and during conditioning of the bed where air can oxidize reactive groups.

Powder x-ray diffraction (Fig. 5B) and N₂ sorption isotherms (Fig. 5C) are shown after a week of exposure to 150°C steam. CALF-20 was also tested for retention of structure and porosity (figs. S12 and S13) after treatment with 20 parts per million (ppm) SO₂ and 100 ppm NO_x at 20°C in separate experiments. We subjected CALF-20 to a real flue gas stream (50°C, flow of 100 cm³ min⁻¹) from natural gas combustion containing 7.3% H₂O, 7.1% O₂, 147 ppm CO, 78 ppm NO, and 13 ppm NO₂ (see supplementary materials, fig. S3, and table S4). Under these flowing flue gas conditions, powdered CALF-20 lost only 1.3% of its capac-

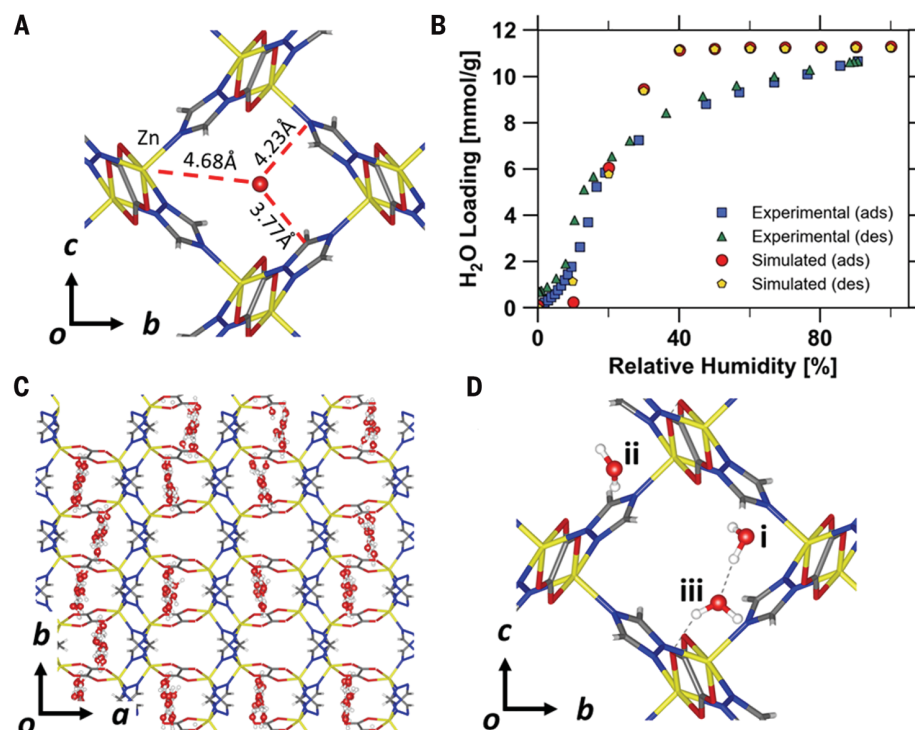


Fig. 3. Most probable CO₂ binding site determined from the single-component CO₂ GCMC simulation at 0.15 atm. (A) Select distances between heavy atoms of the CALF-20 framework and the CO₂ atoms are highlighted. These are the shortest distance between atoms of the framework and any atom of the CO₂ molecule. (B) Experimental and simulated single-component water isotherms at 293 K. Simulated adsorption results refer to the values obtained starting from empty pores, whereas the desorption results refer to the values obtained by starting the simulation with the pores saturated with water. (C) A snapshot from a 20% RH simulation of water in CALF-20. (D) The three most probable H₂O binding sites determined from the single-component water simulation at 20% RH. For (A), (C), and (D), ball-and-stick representations are used for the guest, whereas a tube representation is used for the framework. Atom colors are the same as shown in Fig. 1B.

ity after 6 days, as confirmed by gravimetric CO₂ uptake in a 15/85 mixture of CO₂/N₂. A process demonstration unit using the Svante VelloxTherm process was built on the basis of rotating beds and fast cycles (~1 min) at 0.1 tonne per day CO₂ capacity, and it was deployed to test the CALF-20 lifetime with simulated cement flue gas. This simulated flue gas was generated by enriching real flue gas from a natural gas boiler with pure CO₂ and air to bring CO₂, water, and O₂ concentrations to cement kiln flue gas composition (17% CO₂, 10% O₂, 5% H₂O, balance N₂, at 45°C). The gas analyzer recorded around 60 ppm NO and 12 ppm NO₂ in the generated flue gas that was fed to the CALF-20 beds. The process was continuously tested for over 2000 hours with expected key performance indicators and no appreciable performance loss, as can be seen in fig. S17 (44). Furthermore, the process was able to achieve US Department of Energy target CO₂ purity of 95%.

For large-scale applications, it is important that the scale-up be feasible from an economic and technical viewpoint (45). Analysis of the

cost driver for different MOF syntheses reveals that the costs of the raw materials, especially for the linker and less commonly for the metal, are often prohibitive. In addition, synthetic process conditions can have a substantial impact on the economics—for example, the necessity of high-pressure equipment is not only expensive but also results in costly safety precautions to protect employees and the environment. For CALF-20, none of these disadvantages apply (46). The raw materials are commercially available on a large scale from qualified vendors. Both linkers are low-cost bulk chemicals with large global production capacity (47): 200,000 metric tonnes per annum (MTPA) for oxalic acid, found mainly in pharmaceutical, textile and mining industries; 10,000 MTPA for triazoles, used mainly in the agricultural sector as a building block for azole-based fungicides. In addition, the reaction could be carried out in a water/methanol mixture, where the organic solvent represents <25 wt % with ongoing improvements. These conditions are particularly advantageous from a safety and environmental aspect. Further, in large-scale

The 3D porous metal–organic frameworks based on bis(pyrazinyl)–triazole: structures, photoluminescence and gas adsorption properties†

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Two novel 3D porous metal–organic frameworks, $\{[\text{Zn}_2(\text{bpt})(\text{btc})] \cdot 4\text{H}_2\text{O}\}_n$ (**1**) and $\{[\text{Cd}_2(\text{bpt})(\text{btc})(\text{H}_2\text{O})] \cdot (\text{DMA})_2\}_n$ (**2**) (DMA = *N,N*-dimethylacetamide), were prepared based on 3,5-bis(pyrazinyl)-1,2,4-triazole (Hbpt) with the auxiliary ligand 1,3,5-benzenetricarboxylic acid (H_3btc). Complex **1** could be described as a (3,6)-connected binodal 3D network with rutile topology and Schläfli symbol of $(4.6^2)_2(4^2.6^{10}.8^3)$, where its coordination framework exhibits a high thermal stability up to 420 °C as well as excellent boiling-water resistance. Complex **2** shows a (3,6,8)-connected trinodal 3D framework with a Schläfli symbol of $(4.6^2)_2(4^{14}.6^{12}.8^2)(4^4.6^{10}.8)$, which has not been reported previously. Interestingly, the gas-adsorption investigation discloses that complex **1** possesses a high enthalpy of H_2 adsorption (9.6 kJ mol^{−1}). Intensive solid-state fluorescence spectra of complexes **1** and **2** have also been determined.

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Introduction

Porous metal–organic frameworks (MOFs), emerging as a new class of functional material, have been widely studied not only owing to their topologically interesting architectures but also potential applications in gas storage/separation, optics, catalysis, sensing, and so on.¹ Numerous studies have been reported in the past few years relating to hydrogen storage since hydrogen is regarded as a clean energy that could replace fossil fuels.² It is more convincing that some new materials of MOFs can store a large amount of hydrogen. For example, the hydrogen storage capacities of SNU-77 and MOF-210 are 8.1 wt% (total 11.0 wt%) at 77 K and 90 bar,³ and 8.6 wt% (total 16.7 wt%) at 77 K and 70 bar,⁴ respectively. However, the practical applications of the MOFs as hydrogen storage are very much diminished at ambient temperature. Thus, the isosteric heat (Q_{st}) of H_2 adsorption should be substantially increased to facilitate H_2 adsorption at ambient temperature and pressures. One somewhat successful strategy has been constructing the frameworks with narrow pores to increase the

affinity for H_2 adsorption by using relatively small ligands in the hybrid synthesis.⁵

On the other hand, coordination polymers constructed by N-donor ligands have been widely reported in recent years.⁶ Specifically, 1,2,4-triazole and its derivatives, which combine the coordination geometry of both pyrazoles and imidazoles, are very interesting as bridging ligands.⁷ The 1,2,4-triazole based ligands with pyridyl groups have been employed in the construction of novel architectures.⁸ However, the coordination polymers based on 1H-3,5-bis(pyrazinyl)-1,2,4-triazole (Hbpt), where the 1H-1,2,4-triazole moieties are introduced between the two pyrazine groups, have not been reported. As a versatile tecton, Hbpt contains seven potential N-donors, which can be used as a uni-, bidentate, or tridentate connector to construct diverse metallosupramolecular systems *via* different coordination modes.

Furthermore, it is a rigid angular multifunctional ligand, which is expected to construct many high-dimensional and highly connected coordination frameworks. In order to take advantages of different types of ligands, we chose a polycarboxylic ligand, H_3btc (1,3,5-benzenetricarboxylic acid), as an O-donor auxiliary ligand to synthesize complexes with diverse structures and topologies as well as specific properties.

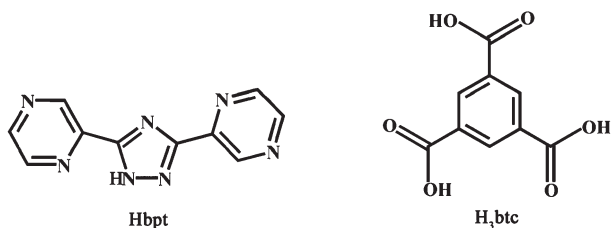
Considering the aforementioned points, we successfully obtained two non-interpenetrated 3D microporous frameworks, $\{[\text{Zn}_2(\text{bpt})(\text{btc})] \cdot 4\text{H}_2\text{O}\}_n$ (**1**) and $\{[\text{Cd}_2(\text{bpt})(\text{btc})(\text{H}_2\text{O})] \cdot (\text{DMA})_2\}_n$ (**2**), derived from Hbpt and H_3btc (Scheme 1) with Zn^{II} / Cd^{II} . The crystal structures, topological analyses, gas adsorption and luminescent properties for both **1** and **2** have been well

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† Electronic supplementary information (ESI) available: The selected bond lengths (Å) and bond angles (°) for complexes **1** and **2**, XRD patterns of complex **2**, and some additional figures. CCDC 926786 and 926787 for complexes **1** and **2**. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c3ce40574e



Scheme 1 The ligands 3,5-bis(pyrazinyl)-1,2,4-triazole (Hbpt) and 1,3,5-benzenetricarboxylic acid (H_3btc).

investigated. More interestingly, complex **1** can be considered as a success of the above approaches to construct a microporous material with narrow pores, which exhibits a remarkable isosteric heat of adsorption for H_2 .

Experimental

Materials and methods

All reactants were reagent grade and used as purchased without further purification. Elemental analyses for C, H, and N were carried out on a German Elementary Vario EL III instrument. The FT-IR spectra were performed on a Nicolet Magna 750 FT-IR spectrometer using KBr pellets in the range of 4000–400 cm^{-1} . Thermogravimetric analyses were recorded on a NETZSCH STA 449C unit at a heating rate of 10 $^{\circ}C\ min^{-1}$ under a nitrogen atmosphere. The power X-ray diffraction (XRD) patterns were collected by a Rigaku DMAX2500 X-ray diffractometer using Cu $K\alpha$ radiation ($\lambda = 0.154\ nm$). Fluorescent analysis of the complexes was performed on a Horiba Jobin–Yvon Fluorolog-3 spectrophotometer analyzer.

Measurement of gas adsorption

N_2 and ambient-pressure H_2 isotherms at 77 K for complexes **1** and **2** were maintained by a liquid nitrogen bath, with pressure ranging from 0 to 1 bar on ASAP (Accelerated Surface Area and Porosimetry) 2020 System, while CO_2 and CH_4 adsorption measurements were done at 273 K/300 K in the same instrument at the same pressure range. The freshly prepared samples of **1** and **2** were soaked in methanol for 3 days to remove free solvates present in the frameworks and then outgassed overnight under vacuum to generate the dehydrated form **1a** and **2a**.

Synthesis of $\{[Zn_2(bpt)(btc)] \cdot 4H_2O\}_n$ (**1**)

A mixture of $Zn(NO_3)_2 \cdot 6H_2O$ (0.6 mmol, 179 mg), Hbpt (0.1 mmol, 22 mg), H_3btc (0.1 mmol, 21 mg), and 10 mL of H_2O –EtOH (1/1, v/v) was sealed in a 23 mL Teflon-lined stainless steel autoclave and heated at 120 $^{\circ}C$ for 3 days. After the mixture was cooled to room temperature, pale yellow block crystals of **1** were obtained with the yield of 89% based on Hbpt. Anal. calcd for $C_{19}H_{17}N_7O_{10}Zn_2$ (fw 634.14): C, 35.98%; H, 2.69%; N, 15.46%. Found: C, 35.34%; H, 2.98%; N, 15.35%. Selected IR data (KBr pellet, cm^{-1}): 3465(s), 3101(s), 2967(w), 1619(s), 1560(s), 1437(s), 1375(s), 1153(w), 1066(w), 936(w), 861(w), 766(m).

Synthesis of $\{[Cd_2(bpt)(btc)(H_2O)] \cdot (DMA)_2\}_n$ (**2**)

Complex **2** was synthesized from $Cd(NO_3)_2 \cdot 4H_2O$ (0.6 mmol, 185 mg), Hbpt (0.1 mmol, 22 mg), H_3btc (0.1 mmol, 21 mg), and 10 mL of H_2O –DMA (1/1, v/v, DMA = *N,N*-dimethylacetamide). The mixture was sealed in a 23 mL Teflon-lined stainless steel autoclave and heated at 120 $^{\circ}C$ for 3 days. After it was cooled down to room temperature, light yellow block crystals of **2** were separated by filtration with the yield of 68% based on Hbpt. Anal. calcd for $C_{27}H_{29}N_9O_9Cd_2$ (fw 848.39): C, 38.22%; H, 3.45%; N, 14.86%. Found: C, 37.98%; H, 3.26%; N, 14.98%. Selected IR data (KBr pellet, cm^{-1}): 3060(w), 2933(w), 1622(s), 1554(s), 1433(s), 1372(s), 1148(w), 1024(w), 866(w), 760(m), 726(m).

X-Ray crystallography

All the structural data of complexes **1** and **2** were collected on a Rigaku Mercury CCD diffractometer equipped with a graphite-monochromated Mo $K\alpha$ radiation ($\lambda = 0.71073\ \text{\AA}$) at room temperature. All of the structures were resolved by the direct method and refined by full-matrix least-squares fitting on F^2 by SHELX-97.⁹ All non-hydrogen atoms were refined with anisotropic thermal parameters. The positions of hydrogen atoms on the organic ligands were generated geometrically and refined using a riding model. In the case of **1**, the hydrogen atoms of O4 were localized on a difference Fourier map and further refined by applying a restraint on the O–H bond lengths. In the case of **2**, hydrogen atoms of O8 could not be determined. In the structure of complex **1**, two water molecules were located in the difference electron density map while two additional highly disordered water molecules were treated with the program PLATON/SQUEEZE. The final chemical formula of **1** was calculated from SQUEEZE results combined with the TGA and elemental analyses data. In addition, the relatively large ADP values for the DMA molecules in **2** might be due to no obvious strong interactions between the free DMA molecules and the host framework. Crystallographic data and structure refinement parameters for complexes **1** and **2** are summarized in Table 1. Selected bond lengths and bond angles are listed in the ESI.[†] More details on the crystallographic studies as well as atomic displacement parameters are given in the ESI[†] as CIF files. CCDC 926786 and 926787 contain the supplementary crystallographic data for this paper.

Results and discussion

Crystal structure of $\{[Zn_2(bpt)(btc)] \cdot 4H_2O\}_n$ (**1**)

A single-crystal X-ray diffraction analysis reveals that complex **1** crystallizes in the tetragonal space group $P4_32_12$ and exhibits a dinuclear Zn_2 -based non-interpenetrated 3D framework. The asymmetric unit is composed of one $Zn(II)$ ion, half of a deprotonated bpt^- ligand, half of a btc^{3-} anion, and two free water molecules in a crystal lattice. As shown in Fig. 1, the Zn center is five-coordinated with two nitrogen atoms (N4, N1B) from two bpt^- ligands and three carboxylate oxygen atoms (O1, O2C, O3C) from different btc^{3-} ligands, resulting in a distorted square pyramid geometry, in which N4, N1B, O2C

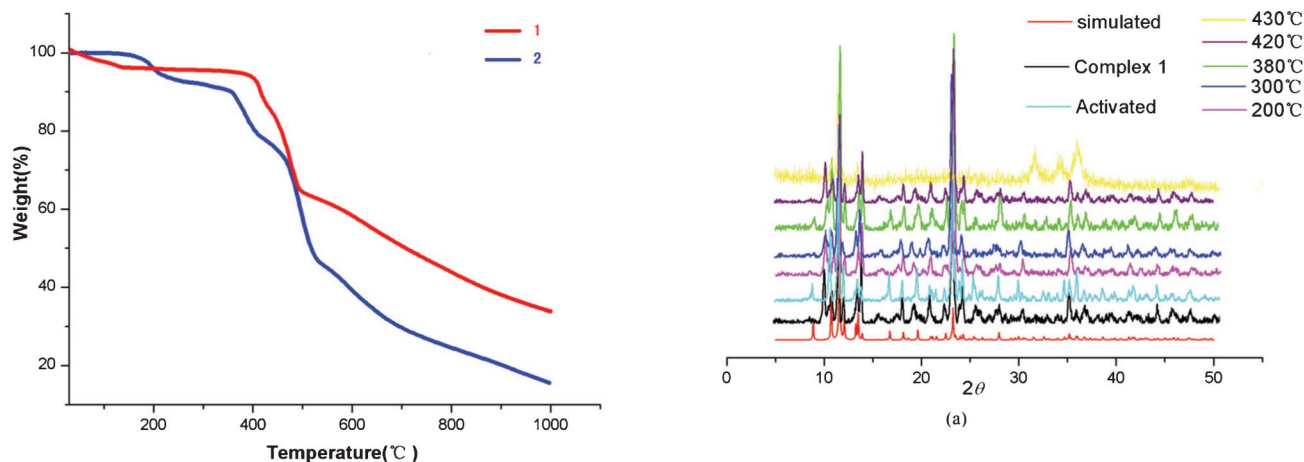


Fig. 7 Thermal gravimetric curves of complexes **1** and **2**.

ESI†). The TGA curve of **2** shows a weight loss of 2.34% below 185 °C, which is ascribed to the release of one coordinated water molecule (calc. 2.12%). From 185 to 400 °C, it loses two free DMA molecules and begins to decompose (calc: 17.61%, found: 17.38%). The TGA of the methanol-exchanged sample (**2-ex**) shows that the guest molecules and the coordinated solvent molecules can be exchanged by methanol (Fig. S1b, ESI†). Subsequently, the samples **1-ex** and **2-ex** are degassed under vacuum, and then the desolvated solids **1a** and **2a** are obtained.

The powder XRD patterns of complexes **1** and **2** in the region of $2\theta = 5\text{--}55^\circ$ are shown in Fig. 8 and S2 (ESI†). The results show that all the peaks displayed in the patterns measured at room temperature closely match those in the simulated patterns from the structure data, indicating the phase purity of the solids. The PXRD pattern of desolvated solid **1a** shows a similar nature to that of the as-synthesized complex **1**. VT-PXRD patterns further indicate that the orders of the framework structure of complex **1** are retained upon complete removal of the guest water molecules. Furthermore, the PXRD patterns of **1** also remain intact upon immersion in water and boiling water, suggesting that no phase transition or framework collapse happens during these treatments. High stability of the coordination frameworks against water as well as the effects of temperature is essential for future practical applications in gas storage. Compared to the PXRD pattern of complex **2**, some main peaks in the PXRD pattern of the desolvated solid **2a** disappear and a few peaks also shift. This result indicates that the host framework of **2** is unstable and may collapse once the guest solvent molecules are removed.

Adsorption properties

Complexes **1** and **2** were subjected to gas adsorption studies. The as-synthesized compounds **1** and **2** were activated by immersing in methanol (MeOH) for 3 days and then outgassed overnight under vacuum to generate the dehydrated forms **1a** and **2a**. For **1a**, a gas adsorption curve is shown in Fig. 9. However, N_2 adsorption of **2a** appeared to fail since the framework collapsed to form a nonporous structure once the

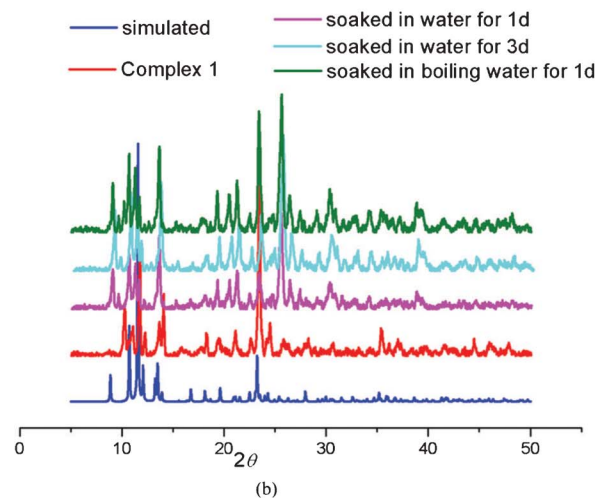


Fig. 8 (a) Variable-temperature PXRD patterns for **1**; (b) PXRD patterns of **1** obtained by different treatment in atmospheric water or boiling water for various durations.

guest solvent molecules were removed under vacuum.¹³ Thus, we merely discuss the details of adsorption properties for **1a** as follows.

The permanent porosity of **1a** was established by reversible gas sorption experiments. The N_2 adsorption isotherm at 77 K exhibits typical type I adsorption behavior, verifying its microporous nature. As shown in Fig. 9a, the maximum N_2 uptake is $100 \text{ cm}^3 \text{ g}^{-1}$ at 77 K, corresponding to a BET surface area of $370.2 \text{ m}^2 \text{ g}^{-1}$. The pore volume calculated from the maximum N_2 adsorption is $0.156 \text{ cm}^3 \text{ g}^{-1}$. By analyzing the crystal structure of **1a**, the internal diameter of the aperture is $4.1 \text{ \AA}$, which is too small to incorporate more N_2 molecules (the kinetic diameter for N_2 , $3.65 \text{ \AA}$). Moreover, there is only a one-dimensional channel system in the framework, and there are no additional channels along the crystallographic *a* and *b* axes in the case of the dehydrated framework of **1**. Therefore, at a low measuring temperature (77 K) the window of the channels is blocked by the entering N_2 molecules, which prevents other N_2 molecules from entering the pores.

H_2 adsorption isotherms were recorded under atmospheric pressures, which also showed type I behavior with good

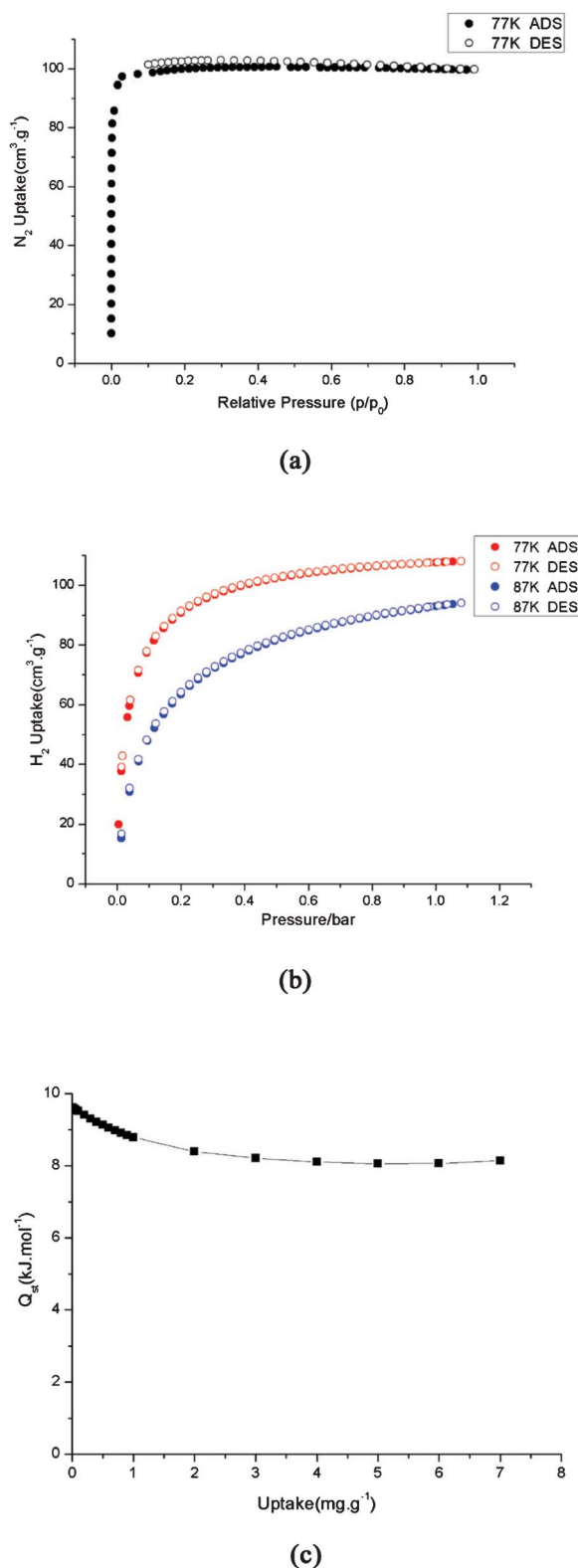


Fig. 9 (a) N₂ sorption isotherms of **1a** at 77 K; (b) H₂ adsorption isotherms of **1a** at 77 K and 87 K; (c) coverage dependency of the isosteric heat of adsorption for H₂ in **1a**.

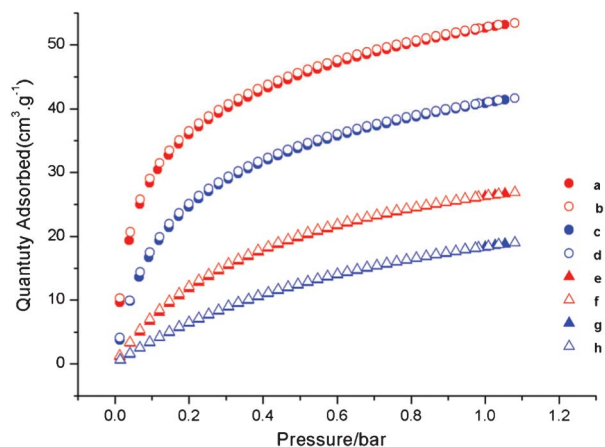


Fig. 10 The sorption isotherms of **1a**: CO₂ adsorption (a) and desorption (b) at 273 K; CO₂ adsorption (c) and desorption (d) at 300 K; CH₄ adsorption (e) and desorption (f) at 273 K; CH₄ adsorption (g) and desorption (h) at 300 K.

reversibility and no hysteresis. At 77 K and 1 bar, it had fully reversible uptake of 108 cm³ g⁻¹ (0.91 wt%), a value closing to those of recently reported MOFs at the same condition.¹⁴ To quantitatively compare their affinities to H₂ molecules, a second H₂ adsorption isotherm was measured at 87 K. The heat (Q_{st}) of H₂ adsorption is calculated by the Clausius–Clapeyron equation and the Q_{st} at zero coverage is estimated to be 9.6 kJ mol⁻¹ (Fig. 9c), higher than those of many famous porous materials, such as MOF-177 (4.4 kJ mol⁻¹), MOF-5 (5.2 kJ mol⁻¹), HKUST-1 (6.6 kJ mol⁻¹), or IRMOF-11 (9.1 kJ mol⁻¹).¹⁵ According to the nature of the pore structure, we consider that the high H₂ binding affinity observed in **1a** can be attributed to its relatively smaller pore size (the kinetic diameter for H₂, 2.89 Å).

To further explore its potential properties in gas sorption under ambient conditions, the adsorption isotherms of CO₂ and CH₄ for **1a** at 273 K and 300 K were measured (Fig. 10). The CO₂ uptake values at 1 bar are 53 cm³ g⁻¹ at 273 K and 42 cm³ g⁻¹ at 300 K. Coverage-dependent CO₂ adsorption enthalpies of **1a** were calculated according to the Clausius–Clapeyron equation. It shows that Q_{st} exhibits a maximum of 28.7 kJ mol⁻¹ at initial coverage (Fig. S3a in the ESI†).

The high Q_{st} for CO₂ benefited from the small channel size of **1**, because the framework–CO₂ interactions can be augmented in small pores. In addition, CH₄ adsorption was also measured at 273 and 300 K under 1 bar. The maximal CH₄ uptakes for **1a** are 27 cm³ g⁻¹ at 273 K and 19 cm³ g⁻¹ at 300 K. The Q_{st} of CH₄ adsorption at zero surface coverage is 20.5 kJ mol⁻¹ (Fig. S3b in the ESI†).

Photoluminescence properties

The photoluminescence of d¹⁰ metal complexes has been attracting intensive research interest, owing to their potential applications in chemical sensors, photochemistry and electroluminescent (EL) displays.¹⁶ Crystal photographs of complex **1** show that it exhibits strong blue emission under UV irradiation (Fig. S4 in the ESI†). The result can be further confirmed by the luminescent spectra. The solid-state emis-

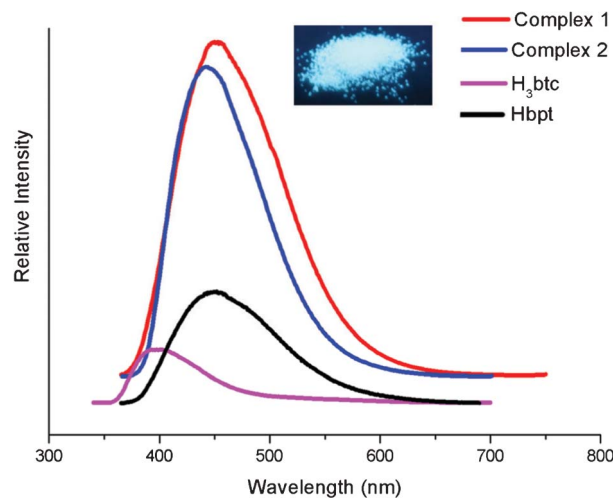


Fig. 11 Solid-state emission spectra of the free ligands and complexes **1** and **2** at room temperature. Inset: crystal photograph of **1** under UV irradiation.

sion spectra of complexes **1–2** with the ligand Hbpt and the O-donor auxiliary ligand H₃btc have been investigated at room temperature. As shown in Fig. 11, complexes **1** and **2** exhibit strong blue emission bands at 453 nm and 443 nm upon excitation at 350 nm, respectively, whereas the Hbpt and H₃btc ligands display very weak photoluminescence with emission maxima at 452 nm and 400 nm upon excitation at 350 nm. The emissions of **1** and **2** can probably be attributed to the interligand (π – π^*) charge transition.^{17,18} In addition, the fluorescent intensity for the coordination polymers is much stronger compared to that of the free ligands, possibly resulting from the enhancement of rigidity of the ligand after the coordination effect and the reduction of the loss of energy by radiation-less decay or vibrational motion.¹⁹ These observations indicate that complexes **1** and **2** may be considered as promising candidates for the exploration of fluorescent materials.

Conclusion

Two novel 3D microporous MOFs based on Hbpt and H₃btc ligands have been synthesized by using solvothermal method. Both of them are non-interpenetrated 3D frameworks with 1D channels. Topologically, complex **1** is a binodal (3,6)-connected three-dimensional network with rutile topology, while complex **2** is a (3,6,8)-connected three-nodal net with the Schläfli symbol of (4.6²)₂(4¹⁴.6¹².8²)(4⁴.6¹⁰.8). It is worth noting that complex **1** with small channels has unusual high stability and exhibits a remarkable heat of H₂ adsorption (9.6 kJ mol^{−1}), which may be expected for materials finding useful applications in hydrogen storage. In addition, the results of photoluminescent measurements indicate that **1** and **2** may be potential blue materials.

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Notes and references

- (a) O. M. Yaghi, M. O'Keeffe, N. W. Ockwig, H. K. Chae, M. Eddaoudi and J. Kim, *Nature*, 2003, **423**, 705; (b) S. Kitagawa, R. Kitaura and S. Noro, *Angew. Chem., Int. Ed.*, 2004, **43**, 2334; (c) S. R. Batten, *CrystEngComm*, 2001, **3**, 67; (d) S. R. Batten and R. Robson, *Angew. Chem., Int. Ed.*, 1998, **37**, 1460; (e) G. Férey, *Chem. Soc. Rev.*, 2008, **37**, 191; (f) M. Dinca, A. Dailly and J. R. Long, *Chem.–Eur. J.*, 2008, **14**, 10280; (g) Y. B. Zhang, W. X. Zhang, F. Y. Feng, J. P. Zhang and X. M. Chen, *Angew. Chem., Int. Ed.*, 2009, **48**, 5287; (h) Z. Y. Li, G. S. Zhu, G. Q. Lu, S. L. Qiu and X. D. Yao, *J. Am. Chem. Soc.*, 2010, **132**, 1490; (i) B. J. Holliday and C. A. Mirkin, *Angew. Chem., Int. Ed.*, 2001, **40**, 2022; (j) J. J. Perry IV, V. C. Kravtsov, G. J. McManus and M. J. Zaworotko, *J. Am. Chem. Soc.*, 2007, **129**, 10076; (k) L. Ma, A. Jin, Z. Xie and W. Lin, *Angew. Chem., Int. Ed.*, 2009, **48**, 9905; (l) J. J. Jiang, L. Li, M. H. Lan, M. Pan, A. Eichhofer, D. Fenske and C. Y. Su, *Chem.–Eur. J.*, 2010, **16**, 1841; (m) S. C. Sahoo, T. Kundu and R. Banerjee, *J. Am. Chem. Soc.*, 2011, **133**, 17950; (n) M. Yoon, K. Suh, H. Kim, Y. Kim, N. Selvapalam and K. Kim, *Angew. Chem., Int. Ed.*, 2011, **50**, 7870; (o) N. C. Jeong, B. Samanta, C. Y. Lee, O. K. Farha and J. T. Hupp, *J. Am. Chem. Soc.*, 2012, **134**, 51.
- (a) L. Schlapbach and A. Züttel, *Nature*, 2001, **414**, 353; (b) D. C. Zhong, J. B. Lin, W. G. Lu, L. Jiang and T. B. Lu, *Inorg. Chem.*, 2009, **48**, 8656.
- H. J. Park, D. W. Lim, W. S. Yang, T. R. Oh and M. P. Suh, *Chem.–Eur. J.*, 2011, **17**, 7251.
- H. Furukawa, N. Ko, Y. B. Go, N. Aratani, S. B. Choi, E. Choi, A. O. Yazaydin, R. Q. Snurr, M. O'Keeffe, J. Kim and O. M. Yaghi, *Science*, 2010, **329**, 424.
- Z. Hulvey, E. H. L. Falcao, J. Eckert and A. K. Cheetham, *J. Mater. Chem.*, 2009, **19**, 4307.
- (a) X. L. Qi, R. B. Lin, Q. Chen, J. B. Lin, J. P. Zhang and X. M. Chen, *Chem. Sci.*, 2011, **2**, 2214; (b) L. Luo, P. Wang, G. C. Xu, Q. Liu, K. Chen, Y. Lu, Y. Zhao and W. Y. Sun, *Cryst. Growth Des.*, 2012, **12**, 2634; (c) R. Natarajan, G. Savitha, P. Dominiak, K. Wozniak and J. N. Moorthy, *Angew. Chem., Int. Ed.*, 2005, **44**, 2115; (d) P. Metrangolo, F. Meyer, T. Pilati, D. M. Proserpio and G. Resnati, *Chem.–Eur. J.*, 2007, **13**, 5765; (e) J. S. Hu, L. Qin, M. D. Zhang, X. Q. Yao, Y. Z. Li, Z. J. Guo, H. G. Zheng and Z. L. Xue, *Chem. Commun.*, 2012, **48**, 681.
- (a) J. G. Haasnoot, *Coord. Chem. Rev.*, 2000, **200–202**, 131; (b) M. Du, X. J. Jiang and X. J. Zhao, *Inorg. Chem.*, 2006, **45**, 3998; (c) J. P. Zhang, Y. Y. Lin and X. C. Huang, *Cryst. Growth Des.*, 2006, **6**, 519; (d) Y. B. Dong, H. Y. Wang, J. P. Ma and R. Q. Huang, *Cryst. Growth Des.*, 2005, **5**, 789; (e) J. P. Zhang, Y. Y. Lin, X. C. Huang and X. M. Chen, *Chem. Commun.*, 2005, 1258; (f) M. Du, X. J. Jiang and X. J. Zhao, *Chem. Commun.*, 2005, 5521.